

Reconstructing frustrated signs from amplitudes with fixed-node and Krylov updates

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(Dated: October 3, 2026)

We separate the sign and positive-amplitude parts of a real many-body wave function, $\psi(x) = s(x)a(x)$ with $a(x) = |\psi(x)| > 0$, in the frustrated square-lattice spin-1/2 J_1 - J_2 Heisenberg model. For a fixed amplitude, one shifted Krylov step gives $s'(x) = s(x) \text{sgn}[T - r_s(x)]$, where $r_s(x) = (H\psi)(x)/\psi(x)$. At $J_2/J_1 = 0.5$ this reduces the exact 4×4 probability weight on incorrectly signed configurations from 1.27×10^{-2} to 3.41×10^{-4} . On 6×6 , using exactly the same neural-network amplitude, the reconstructed signs give the same energy as the original neural-network signs within sampling uncertainty. Conversely, lattice fixed-node projection improves the positive amplitude while keeping the signs fixed; an exact 4×4 fixed-node/Krylov loop reaches the exact ground-state signs. The reconstructed signs remain useful on 8×8 . The remaining scaling problem is to infer from fixed-node Monte Carlo walkers a positive amplitude function that can be evaluated on arbitrary configurations. Maximum likelihood targets $a_{\text{new}}^2 \propto a\phi_{\text{FN}}$, and stochastic reconfiguration produces an update that generalizes between independent walker samples.

I. INTRODUCTION

The Monte Carlo sign problem is severe in frustrated spin systems [1, 2]. We use the square-lattice J_1 - J_2 Heisenberg model as a benchmark. At $J_2 = 0$ its ground-state signs obey the Marshall rule [3]; near $J_2/J_1 = 1/2$ that rule fails and accurate variational states become nontrivial [4, 5].

We ask whether signs and positive amplitudes can be improved separately. For a real wave function we write

$$\psi(x) = s(x)a(x), \quad s(x) = \pm 1, \quad a(x) = |\psi(x)| > 0. \quad (1)$$

Given a , a single Krylov step updates the signs. Given s , lattice fixed node gives a sign-free positive-amplitude problem [6]. Our calculations first test these two operations separately and then close the feedback loop.

II. MODEL AND METHOD

We study

$$H = J_1 \sum_{\langle ij \rangle} \mathbf{S}_i \cdot \mathbf{S}_j + J_2 \sum_{\langle\langle ij \rangle\rangle} \mathbf{S}_i \cdot \mathbf{S}_j, \quad (2)$$

on periodic $L \times L$ lattices with $J_1 = 1$ in the $S^z = 0$ sector. For a spin configuration x , the Marshall rule is $s_M(x) = (-1)^{N_{\downarrow}^A(x)}$ up to an overall sign, where $N_{\downarrow}^A$ counts down spins on one checkerboard sublattice.

A. One-step sign update

For $\psi = sa$, define the local energy

$$r_s(x) = \frac{(H\psi)(x)}{\psi(x)}. \quad (3)$$

Because

$$[(T - H)\psi](x) = [T - r_s(x)]\psi(x), \quad (4)$$

the sign of the shifted Krylov vector is

$$s_K(x) = s(x) \text{sgn}[T - r_s(x)]. \quad (5)$$

Only this sign is kept; the Krylov amplitude factor $|T - r_s(x)|$ is discarded. Thus the method is not repeated Lanczos multiplication. For exact systems, T is chosen to minimize the fixed-amplitude variational energy. For sampled systems, T is obtained from a two-cluster split of the $r_s(x)$ values and therefore does not use target sign labels.

When the exact ground state $\psi_0 = s_0 a_0$ is available, we fix the irrelevant overall sign and report directly the wrong-sign probability weight

$$w_s = \sum_{x: s(x) \neq s_0(x)} a_0(x)^2. \quad (6)$$

B. Fixed-node amplitude update

For a trial wave function ψ_T , an off-diagonal matrix element connecting x and y is sign violating when $H_{xy}\psi_T(y)/\psi_T(x) > 0$. Lattice fixed node removes such off-diagonal terms and adds their contribution to the diagonal [6]. After importance sampling, the resulting Hamiltonian is sign free and keeps the signs of ψ_T fixed.

Let $\phi_{\text{FN}}(x) > 0$ be the positive ground-state amplitude obtained from this fixed-node Hamiltonian. In an exact small-system calculation, ϕ_{FN} can be inserted directly into Eq. (5). In a walker calculation, however, the stationary sampled distribution is

$$f(x) \propto a(x)\phi_{\text{FN}}(x). \quad (7)$$

If a positive model is trained by maximum likelihood with $p_\theta(x) \propto a_\theta(x)^2$, the exact maximum-likelihood target is

$$a_{\text{new}}^2(x) \propto a(x)\phi_{\text{FN}}(x). \quad (8)$$

Since $a_{\text{new}}(x) > 0$, this is equivalent to taking the geometric mean of the current and fixed-node amplitudes.

The 4×4 exact calculations use the full $S^z = 0$ Hilbert space of 12870 states. For 6×6 and 8×8 we use a previously optimized translationally invariant vision-transformer neural quantum state (ViT) at $J_2/J_1 = 0.5$. In the sign-only benchmarks, its positive amplitude $|\psi_{\text{ViT}}(x)|$ is held fixed. The 8×8 fixed-node calculations use a walker population $M = 128$ and projection length $\beta = 1.2$.

III. RESULTS

A. One-step proof of concept

The simplest test keeps the positive amplitude fixed and changes only the signs. On exactly tractable clusters we can therefore measure both the sign error and the corresponding energy error directly.

At $J_2/J_1 = 0.5$, the $N = 16$ wrong-sign probability decreases from 1.27×10^{-2} to 3.41×10^{-4} , a factor 37.4, while the relative energy error falls from 2.19×10^{-2} to 1.04×10^{-3} , a factor 21.0. On the independent $N = 20$ exact cluster the corresponding reductions are 49.5 and 24.1. The one-step improvement is therefore not peculiar to the 4×4 cluster.

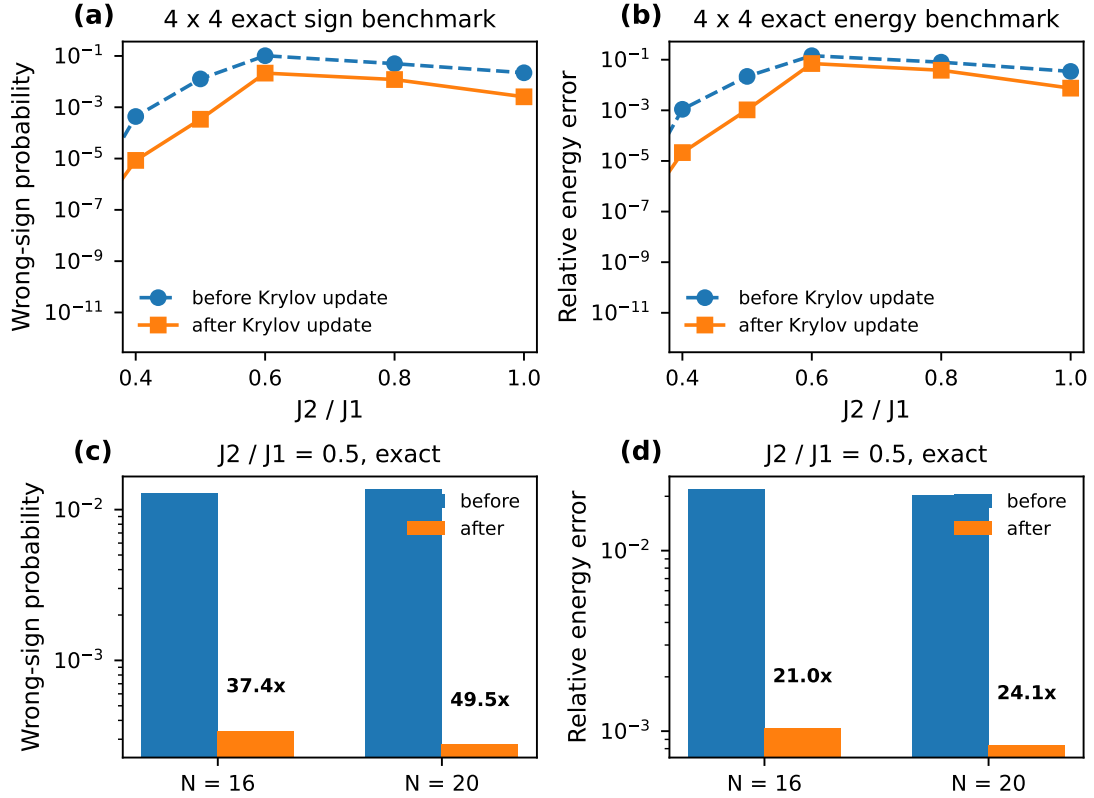


FIG. 1. One-step proof of concept. (a,b) Exact 4×4 sweep of the wrong-sign probability and relative variational energy error $\epsilon_{\text{rel}} = (E - E_0)/|E_0|$, keeping the exact positive amplitude fixed. (c,d) The same before/after comparison at $J_2/J_1 = 0.5$ on two exact periodic clusters: the $N = 16$ square and an $N = 20$ skew torus. The wrong-sign probability is reduced by factors 37.4 and 49.5, while the energy error is reduced by factors 21.0 and 24.1, respectively. For the 4×4 sweep, the initial sign rule is Marshall through $J_2/J_1 = 0.6$ and the corresponding stripe reference at $J_2/J_1 \geq 0.8$. Exact zeros are shown at 10^{-12} on logarithmic axes.

B. Larger-system validation

We next remove the exact-sign benchmark. On 6×6 we use the same ViT positive amplitude with three sign choices: Marshall, one-step Krylov, and the original ViT signs.

The 6×6 energies per site are

$$\begin{aligned} e_{\text{Marshall}} &= -0.4981534, \\ e_{\text{Krylov}} &= -0.5033337, \\ e_{\text{ViT}} &= -0.5034504. \end{aligned} \tag{9}$$

The paired difference $e_{\text{Krylov}} - e_{\text{ViT}} = 1.17(287) \times 10^{-4}$ is unresolved, whereas Krylov improves over Marshall by 4.79×10^{-3} per site. The larger-system data therefore support the same conclusion as the exact benchmark: the one-step sign update captures the energetically relevant sign structure.

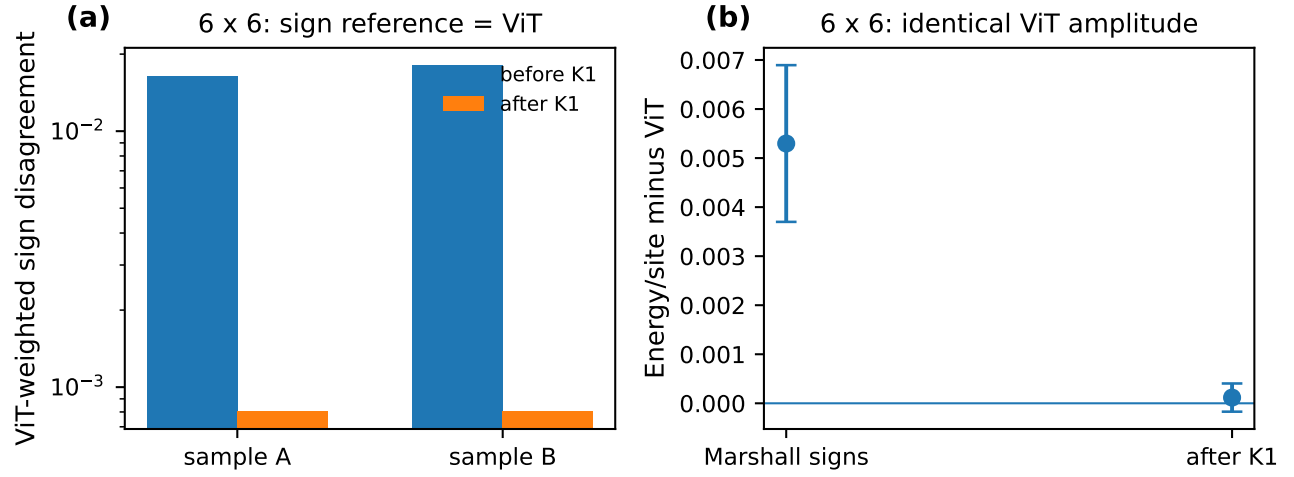


FIG. 2. Larger-system validation on 6×6 . Here the optimized ViT state is a reference rather than an exact ground state. (a) On two independent samples, the $|\psi_{\text{ViT}}|^2$ -weighted sign disagreement with the ViT signs before and after one Krylov update. (b) Fixed-amplitude energy per site relative to the ViT-sign energy, using exactly the same ViT positive amplitude. The Krylov-ViT difference is statistically unresolved, while the Marshall-ViT difference is clearly positive.

C. Closing the fixed-node-Krylov loop

We now test whether the positive amplitude can also be updated. We target $J_2/J_1 = 0.5$ but initialize with the exact $J_2 = 0$ positive amplitude and Marshall signs. At every iteration we construct the current fixed-node Hamiltonian, diagonalize it exactly to obtain its positive ground-state amplitude $\phi_{\text{FN}}(x)$, use that amplitude for the next state, and then recompute the signs with Eq. (5). No target-ground-state information enters either update; exact diagonalization of the physical Hamiltonian is used only to evaluate the plotted errors.

On $N = 16$, the wrong-sign probability falls from 1.27×10^{-2} to 8.85×10^{-7} after twenty iterations and reaches zero to numerical precision by iteration 100. The final amplitude fidelity is 0.9999946 and the trial-state relative energy error is 3.08×10^{-6} . The independent $N = 20$ cluster shows the same behavior: by iteration 40 the wrong-sign probability is 2.55×10^{-7} , the amplitude error is 8.15×10^{-5} , and the relative energy error is 1.41×10^{-5} . Thus the exact feedback mechanism survives beyond the 4×4 cluster.

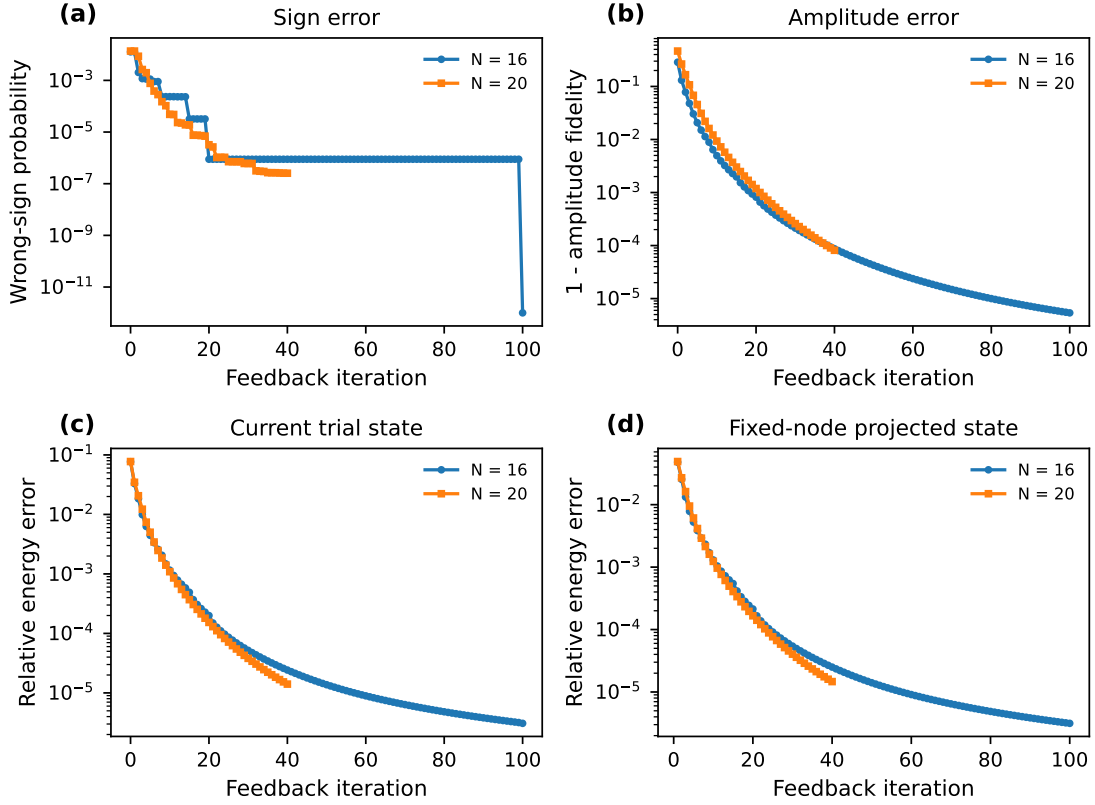


FIG. 3. Exact fixed-node/Krylov feedback loop at $J_2/J_1 = 0.5$ on two periodic clusters, $N = 16$ and $N = 20$. Each iteration exactly diagonalizes the current fixed-node Hamiltonian to update the positive amplitude and then applies one Krylov sign update. (a) Wrong-sign probability. (b) Amplitude error $1 - F_a$, with $F_a = |\langle a|a_0 \rangle|^2$. (c) Relative energy error $\epsilon_{\text{rel}} = (E - E_0)/|E_0|$ of the updated trial state. (d) The same energy error for the fixed-node projected state.

D. Scaling to 8×8

At $M = 128$, the mean total fixed-node energies from two independent walker populations are

$$E_{\text{Krylov}} = -31.88542, \quad E_{\text{Marshall}} = -31.67748. \quad (10)$$

Using the Krylov signs gives $E/N = -0.498210$, compared with -0.494961 for Marshall signs. Figure 4 places this in the range of high-accuracy same-geometry benchmarks. With only two populations we treat their spread as a reproducibility scale rather than a precision error bar.

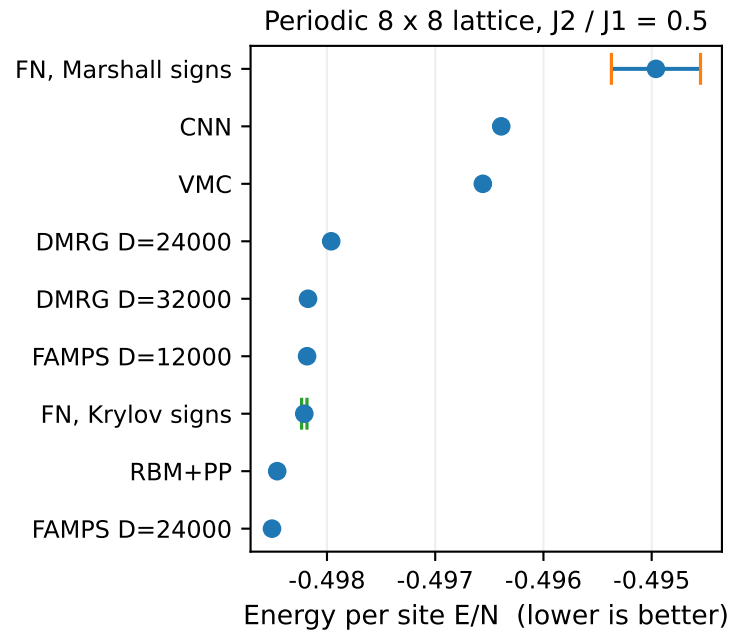


FIG. 4. Energy per site on the periodic 8×8 lattice at $J_2/J_1 = 0.5$. “FN” denotes lattice fixed node. The two FN entries use Marshall signs or the one-step Krylov signs. Short ticks show the two independent walker populations. Literature values are from Ref. [7].

E. Learning the positive amplitude from walkers

For sampled fixed-node walkers, Eq. (8) defines the amplitude learning target. Its exact 4×4 behavior and the first scalable 8×8 learning step are shown in Fig. 5.

On 8×8 , ordinary maximum-likelihood gradients from two independent walker samples have Euclidean cosine -0.853 , so a standard gradient step does not transfer between samples. With stochastic reconfiguration,

$$\delta\theta = (S + \lambda I)^{-1} g_1, \quad (11)$$

and $\lambda = 1$, the directional derivative on the second sample is $+3.85$. A finite step $\eta = 0.01$ gives a positive independent-sample log-likelihood change of 0.0156 with effective sample size $3236/4096$. This establishes a transferable first amplitude update, but not yet convergence of repeated 8×8 feedback iterations.

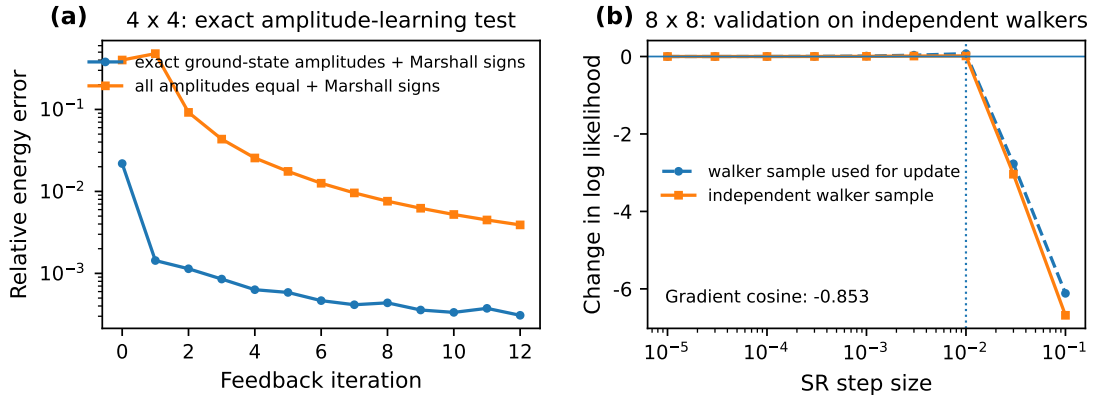


FIG. 5. Learning the positive amplitude. (a) Exact 4×4 test of Eq. (8), plotted as relative energy error $\epsilon_{\text{rel}} = (E - E_0)/|E_0|$. The two curves start from the exact target amplitudes or from equal positive amplitudes, in both cases with Marshall signs. (b) Stochastic-reconfiguration (SR) step-size scan on 8×8 : the update is fitted to one fixed-node walker sample and evaluated on an independent sample.

IV. DISCUSSION AND OUTLOOK

The data separate the two tasks. With a good positive amplitude, the dominant frustrated sign correction is captured by a threshold in the local energy. Fixed node solves the complementary positive-amplitude problem at fixed signs, and the exact 4×4 loop shows that the two updates can close.

The unresolved large-system problem is learning that positive amplitude from projector walkers. SR makes one 8×8 update generalize between independent walker samples, but repeated updates still require fixed-node energy validation. The method also requires a useful initial amplitude and should not be confused with repeated bare Krylov iteration; the direct second step is worse on 6×6 .

Appendix A: Direct second Krylov step

On 6×6 , retaining the full first Krylov amplitude gives $a_1(x) = |T_1 - r_0(x)|a_0(x)$ with coefficient of variation about 0.18. A second unsupervised threshold changes only 0.8–2.0% of the $|\psi_{\text{VIT}}|^2$ probability weight but raises the fixed-amplitude energy by 0.65–0.96 in total energy. The feedback construction therefore restores or re-solves the amplitude before another sign update.

Appendix B: Statistical convention

Exact-diagonalization data are deterministic up to eigensolver precision. For the 6×6 fixed-amplitude comparison we use correlation-robust between-chain standard errors. For 8×8 fixed-node data with two independent walker populations, error bars denote half the population difference and are treated only as reproducibility scales.

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